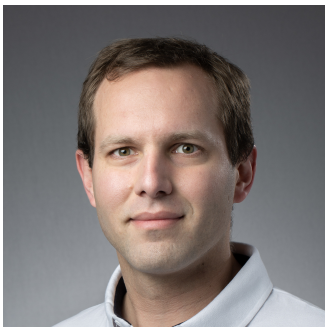


Péter Juhász, PhD

Quantitative Researcher

CONTACT

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- LinkedIn
- shepherd92

EDUCATION

2022 – 2025 PhD in Mathematics
Aarhus University, Denmark

2019 – 2021 MBA in Finance
Budapest University of Technology and Economics, Hungary

2013 – 2016 MSc in Physics
Budapest University of Technology and Economics, Hungary

2010 – 2013 BSc in Physics
Budapest University of Technology and Economics, Hungary

SKILLS

Modeling: Stochastic Models, Bayesian Statistics, Probability, Data Analysis

Programming: Machine Learning, Python, C++, SQL, Git, Monte Carlo Simulation

LANGUAGES

English

German

Hungarian

PROFILE

Quantitative researcher with a PhD in Mathematics, combining expertise in stochastic modeling, statistical inference, and machine learning with industrial experience in developing predictive systems. Experienced in large-scale time-series analysis, uncertainty quantification, and rigorous model validation, with strong Python/C++ skills and a proven track record of translating research into production software. Currently developing intraday electricity-price models and stochastic-control methods for optimal liquidation at Aarhus University, in collaboration with an energy trading firm.

PROFESSIONAL EXPERIENCE

Quantitative Researcher
Energy Markets & Algorithmic Trading **2026 —**
Aros Commodities & Aarhus University, Denmark

- Developed stochastic models of high-frequency intraday electricity prices for optimal liquidation.
- Modeled market microstructure and auction effects using Hawkes processes and empirically calibrated jump distributions.
- Built memory-efficient Python tools for large-scale quote analysis, maximum-likelihood estimation, and out-of-sample model evaluation.
- Developed stochastic-control methods for optimal liquidation using Hamilton–Jacobi–Bellman equations.

PhD Researcher in Mathematics **2022 — 2025**
Aarhus University, Denmark

- Developed stochastic models and established limit theorems characterizing their distributional and asymptotic behavior.
- Designed statistical tests to assess model assumptions and agreement with empirical network data.
- Implemented high-performance Monte Carlo algorithms in Python and C++ to investigate finite-sample behavior and validate theoretical predictions.

CERTIFICATIONS

2021 **Software Architect** ISAQB

2021 **Software Designer** Bosch

AWARDS, GRANTS

2022 **PhD Fellowship**
Danish Data Science Academy
1 890 000 DKK

2021 **Excellence Award**
Hungarian National Bank
350 000 HUF

Machine Learning Researcher
& Software Architect **2018 — 2022**
Bosch Group, Hungary

- Patented a Gaussian regression-based trajectory prediction model, improving predictive accuracy by $3\times$ and uncertainty calibration by $10\times$ over production baselines.
- Built stochastic forecasting and uncertainty quantification models for safety-critical, real-time decision-making in automated driving.
- Analyzed more than 20 000 hours of driving data to develop and validate probabilistic forecasting models for safety-critical decision-making.
- Translated research prototypes into scalable production software with explicit reliability requirements.
- Served as technical lead for an 11-person team developing, validating, and integrating automated-driving prediction models.

Software Engineer
Performance Analytics **2016 — 2018**
Ericsson Telecommunications, Hungary

- Analyzed response-time distributions and system performance to improve load balancing in large-scale distributed systems.
- Designed and implemented performance-critical algorithms for high-throughput network routing.

PATENTS

- P. Juhász et al. **2025**
  Method and apparatus for processing data associated with at least one ultrasonic sensor
- P. Juhász. **2021**
  Method and device for predicting the trajectory of a traffic participant, and sensor system

SELECTED PUBLICATIONS

- P. Juhász. **2025**
  PhD Thesis: Stochastic Models of Higher-Order Networks - Point Processes and Topological Data Analysis
- C. Hirsch, B. Jahnel, and P. Juhász. **2025**
  Functional limit theorems for edge counts in dynamic random connection hypergraphs
- P. Juhász. **2021**
  Information propagation in stochastic networks
- P. Juhász, J. Stéger, D. Kondor, and G. Vattay. **2018**
  A Bayesian approach to identify Bitcoin users